

Adapt App

Build binaries & publish to GitHub

This guide explains how to package the Adapt App as a standalone executable with PyInstaller, for Linux and Windows, and how to upload both builds to GitHub Releases so friends can download them without installing Python.

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Part 1 - Linux binary (PyInstaller)

You develop on Linux (e.g. Fedora). The goal is a folder your users can extract and run without Python installed. Adapt App entry point: `python -m src.core.main`

1.1 Prerequisites

- Python 3.11+ and git
- Project folder cloned from GitHub
- System libraries for pygame/SDL (Fedora example below)

```
sudo dnf install python3-pip python3-devel SDL2 SDL2_image SDL2_mixer SDL2_ttf
cd "Adapt App"
python -m venv venv
source venv/bin/activate
pip install -e ".[dev]"
```

1.2 Create adapt.spec (one-time)

The old main.spec points at Backend/Core/main.py and is outdated. Create adapt.spec in the project root:

```
# adapt.spec - save in project root
block_cipher = None
from PyInstaller.utils.hooks import collect_all
pg_datas, pg_binaries, pg_hidden = collect_all('pygame')

a = Analysis(
    ['src/core/main.py'],
    pathex=['.'],
    binaries=pg_binaries,
    datas=[('NeuroMods', 'NeuroMods')] + pg_datas,
    hiddenimports=['pandas', 'fsrs', 'pygame', 'sqlite3'] + pg_hidden,
    noarchive=False,
)
pyz = PYZ(a.pure)
exe = EXE(
    pyz, a.scripts, [], exclude_binaries=True,
    name='AdaptApp', console=True,
)
coll = COLLECT(exe, a.binaries, a.datas, name='AdaptApp')
```

Note: NeuroMods/ must be bundled - the bar game and flashcard import depend on it. Large image folders make the build big (~hundreds of MB); that is normal.

1.3 Build

```
source venv/bin/activate
pyinstaller adapt.spec --noconfirm
```

Output folder: `dist/AdaptApp/`

Test before uploading:

```
./dist/AdaptApp/AdaptApp
```

1.4 Linux edge cases

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- Missing NeuroMods: if the bar trainer says it cannot find NeuroMods/Bar, check datas= in adapt.spec includes ('NeuroMods', 'NeuroMods').
- Database writes: adaptations.db lives at src/adaptations.db in dev. Frozen builds use sys._MEIPASS (read-only). For production, copy an empty DB to ~/.local/share/AdaptApp/ on first run, or document that users run from a writable folder.
- Wayland / KDE maximize crash: if the bar game segfaults on launch, try SDL_VIDEODRIVER=x11 ./dist/AdaptApp/AdaptApp or run windowed (F11).
- Fonts: install liberation-sans-fonts for readable bar UI text on Fedora.
- One-file vs one-folder: this guide uses one-folder (faster startup, easier to debug). --onefile is possible but slower to start and harder to patch.
- glibc: build on an older distro (or CI with manylinux) if friends use older Linux - binaries linked against too-new glibc won't run elsewhere.

1.5 Package for upload

```
cd dist
tar -czvf AdaptApp-linux-x86_64.tar.gz AdaptApp/
```

Part 2 - Windows binary (for a non-technical friend)

Your friend does not need to understand Python. They only need to build on a Windows PC (or you build in CI) and send you the zip. Give them this checklist verbatim.

2.1 What your friend needs

- Windows 10 or 11
- Internet to download two installers (Python + Git - optional if you send a zip)
- The Adapt App project folder (Download ZIP from GitHub)

2.2 Step-by-step for your friend

Step 1 - Install Python

- Go to <https://www.python.org/downloads/>
- Download Python 3.11 or newer
- Run installer. CHECK "Add python.exe to PATH" at the bottom - important!
- Click Install Now

Step 2 - Open Command Prompt in the project folder

- Unzip Adapt App to Desktop\Adapt App
- In the folder, click the address bar, type cmd, press Enter

Step 3 - Copy and paste these commands one block at a time

```
python -m venv venv
venv\Scripts\activate
pip install -e ".[dev]"
```

Step 4 - Create adapt.spec

Copy the adapt.spec contents from Part 1.2 into a new file named adapt.spec in the project root (Notepad: File > Save As > adapt.spec, save as All Files).

Step 5 - Build

```
pyinstaller adapt.spec --noconfirm
```

Wait several minutes. When done, the app is in dist\AdaptApp\AdaptApp.exe

Step 6 - Test

- Double-click dist\AdaptApp\AdaptApp.exe
- Choose 5 for Bar practice - window should open
- If Windows SmartScreen warns: More info > Run anyway (unsigned exe)

Step 7 - Zip and send to you

- Right-click dist\AdaptApp folder > Send to > Compressed (zipped) folder
- Email / Google Drive / Discord - send AdaptApp-windows.zip

2.3 Windows edge cases

- Antivirus false positives: unsigned PyInstaller exes are often flagged. Build on a clean machine; upload SHA256 checksums on GitHub.
- Missing DLL / pygame: if the exe fails immediately, reinstall with `pip install -e ".[dev]"` and rebuild; ensure `collect_all('pygame')` is in spec.
- Path with spaces: project folder name Adapt App is fine; always quote paths in scripts.

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- Long paths: enable Windows long path support if build fails on deep NeuroMods paths.

Part 3 - Publish Linux & Windows on GitHub

Use GitHub Releases so users download a zip, extract, and run - no git or pip.

3.1 Prepare release assets

- AdaptApp-linux-x86_64.tar.gz (from Part 1.5)
- AdaptApp-windows.zip (from your friend's Part 2)
- Optional: SHA256 checksums text file for both

3.2 Create a release on GitHub

- Open your repo on github.com
- Click Releases (right sidebar) > Create a new release
- Choose a tag, e.g. v0.0.1-alpha - Create new tag on publish
- Release title: Adapt App v0.0.1-alpha
- Describe what's new (bar trainer, customer portraits, etc.)

3.3 Upload binaries

- Drag both archives into Attach binaries
- Publish release

3.4 README snippet for users

```
## Downloads
Get the latest release for your system:
- **Linux**: AdaptApp-linux-x86_64.tar.gz
  tar -xzf AdaptApp-linux-x86_64.tar.gz && ./AdaptApp/AdaptApp
- **Windows**: AdaptApp-windows.zip
  Extract and run AdaptApp.exe
```

3.5 Ongoing releases

- Tag each release (v0.0.2, etc.) - never overwrite old tags
- Build Linux binary yourself; friend rebuilds Windows when you ask
- Or use GitHub Actions later to build both automatically on push tag

3.6 Edge cases on GitHub

- 100 MB limit per file on free GitHub - if NeuroMods makes the zip huge, consider Git LFS or splitting asset packs.
- License: ensure LICENSE allows redistribution of bundled NeuroMods images.
- Do not commit dist/ or build/ - add to .gitignore; only upload Release assets.

Quick reference

```
Linux build: pip install -e ".[dev]" && pyinstaller adapt.spec
Linux run:   ./dist/AdaptApp/AdaptApp
Windows run: dist\AdaptApp\AdaptApp.exe
Dev run:    python -m src.core.main
```